

Campus Canteen (MessQ) – Project Report

Comprehensive Technical & Architectural Overview for Stakeholders & Partners

1. Executive Summary & Purpose

Campus Canteen (MessQ) is a smart campus food ordering and queue management system designed specifically to eliminate cafeteria rush, crowded lines, and ordering chaos in colleges and universities. Students can browse block-wise canteens, inspect real-time crowd levels, place customized orders with dietary choices, and track order preparation in real time. Kitchen staff receive a dedicated management terminal to accept, prepare, and dispatch orders with two-factor email OTP security.

2. Technology Stack & Cloud Infrastructure

Component	Technology	Key Role & Responsibility
Mobile Client	Android · Kotlin 2.x	Native mobile architecture with Coroutines & StateFlow
User Interface	Jetpack Compose (Material3)	100% declarative UI with Amber & Orange design system
Primary Database	Supabase (PostgreSQL)	Relational schema with Row Level Security & Realtime WebSockets
Companion Backend	Firebase Suite	Firestore offline cache, Google Credentials auth, FCM push alerts
Email 2FA Security	Resend API	Dispatches automated 6-digit branded HTML verification codes
Networking & Images	Retrofit, OkHttp3, Coil	REST API calls, network diagnostics, and cached image rendering

System Architecture Diagram

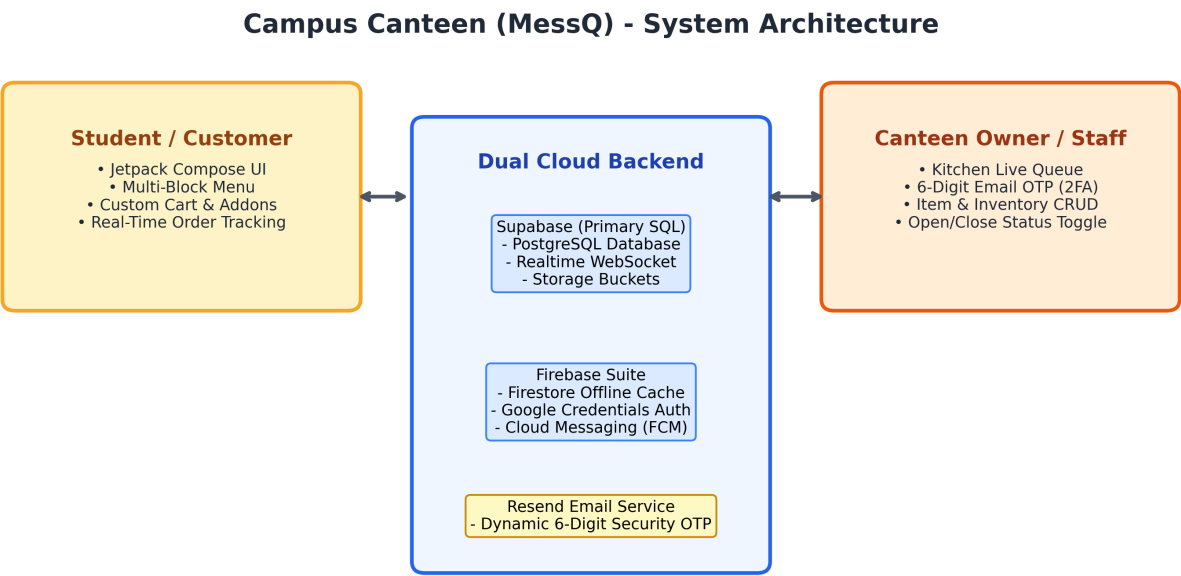


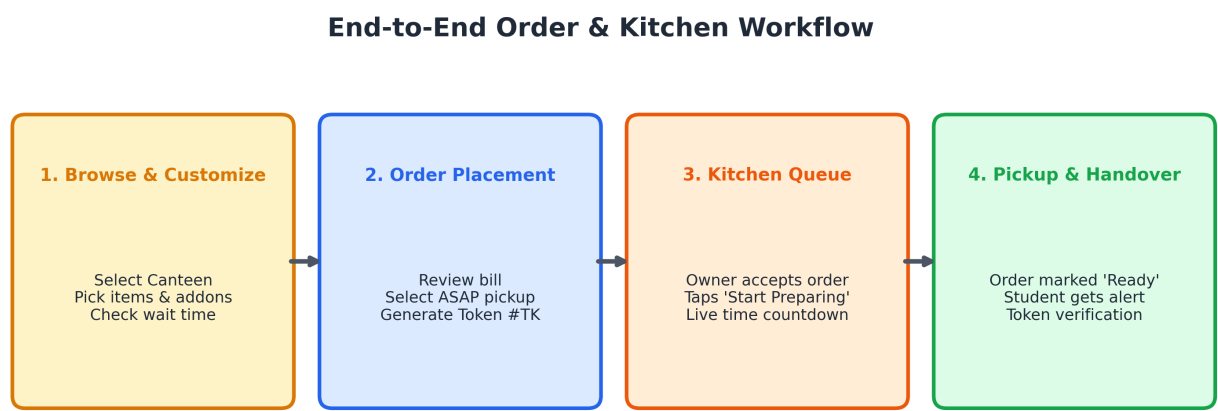
Figure 1: Dual Backend Architecture & Client-Server Connectivity

3. System Roles & Access Matrix

Capability / Feature	Student / Customer	Canteen Owner / Staff	Guest Visitor
Authentication	Email, Google, Reg. No.	Email + 6-digit Resend OTP	Explore mode
Menu & Crowd Rush	Yes (All campus blocks)	Assigned Canteen only	Yes (View-only)
Dish Customization	Bread type, roast, addons	Configure ingredients	View-only
Order Placement	Yes (Generates Token #TK)	No	Requires Sign-In
Live Order Tracking	Yes (4-step visual stepper)	Full kitchen queue view	No
Kitchen Queue Control	No	Accept, Ready, Deliver	No
Menu Item Management	No	Add, edit, in/out of stock	No
Open/Close Broadcast	No	Yes (Toggle with reason)	No
Revenue Reports	No	Daily sales & order metrics	No

4. End-to-End Operational Workflow

Orders follow a strict 4-stage lifecycle ensuring synchronized status updates across both parties:



Real-time sync between Student and Owner views powered by Supabase & Firebase

Figure 2: Four-Stage Real-Time Ordering and Kitchen Execution Flow

5. Complete Screen Catalog (27 Screens)

Category	Screen Name	Key Functions & User Actions
Onboarding & Auth	SplashScreen	Validates session tokens, connectivity, and routes to appropriate entry portal.
Onboarding & Auth	NoInternetScreen	Offline screen with one-tap connection re-validation.
Onboarding & Auth	WelcomeScreen	Landing screen with 'Get Started', 'Sign In', and 'Explore as Guest'.
Onboarding & Auth	ModernSignInScreen	Student credentials + 8-digit Reg No., Course, and 1-tap Google Sign-In.
Onboarding & Auth	OwnerLoginScreen	Kitchen staff portal with credentials authentication + 6-digit email OTP.
Onboarding & Auth	ResetPasswordScreen	Handles deep links (messq://reset-password) for credential updates.
Student Experience	HomeScreen	Multi-block canteen selector, live crowd rush badge, quick order categories.
Student Experience	MenuScreen	Food catalog with Veg/Non-Veg filter, price/time sorting, and search.
Student Experience	FoodDetailScreen	Dish calories, prep minutes, ingredients list, bread/spice options, and addons.
Student Experience	CartScreen	Itemized cart breakdown, customized addons, price calculations, and ASAP pickup scheduling.
Student Experience	CustomerOrderTrackingScreen	Live 4-step progress stepper, estimated wait countdown, and counter location.
Student Experience	OrdersScreen	Split active live orders and past receipts with quick re-order shortcuts.
Student Experience	ProfileScreen	Student registration details, saved favorite foods, and logout action.
Owner Operations	OwnerDashboardScreen	Today's revenue (■), orders count, active queue, and canteen open/close switch.
Owner Operations	OwnerOrdersScreen	Live kitchen queue with status advance: New -> Preparing -> Ready -> Handover.
Owner Operations	ManagelItemsScreen	Menu editor to create dishes, change prices, and toggle in/out of stock.
Owner Operations	InventoryScreen	Stock and raw material inventory tracking with low-quantity warnings.
Owner Operations	OwnerCustomersScreen	Student customer directory with order counts and quick contact access.
Owner Operations	OwnerProfileScreen	Canteen timings, location info, staff emails, and security settings.

6. Architectural Highlights & Key Features

- **Block-Aware Crowd Status:** Calculates live kitchen rush (Fast Service, Moderate Crowd, High Rush) and wait times in minutes based on active queue volume.
- **Contextual Dish Customization:** Supports option selection (Bread type, Roast level, Spice level) paired with paid addons (Extra Cheese, Extra Sauce) updating order totals dynamically.
- **Floating Quick-Cart Capsule:** Non-intrusive animated bottom widget showing newly added items with a one-tap shortcut to view cart.
- **Resend Two-Factor OTP Security:** Kitchen administration is protected by dispatching 6-digit verification codes to the owner's verified email via the Resend API.
- **Canteen Status Broadcast:** Allows owners to pause incoming orders during breaks or kitchen restocking, broadcasting the reason across student screens.